pg-WRS for Interim I

Group 6: Kyle Gothman, Cynthia Corsiga, Grace Seeberger, Aaron Tran, Luis Alarcon, Dylan Hinton, Julia Bacud, Adam Hassan, Quincy Kemany

Revision History

Date	Author(s)	Change

Process

The team held one meeting on Saturday, September 19th, from 2:00-3:00 PM, and held another meeting on Saturday, September 26th, from 2:00-3:00 PM. During the first meeting, the team created blank drafts for each deliverable and assigned portions of the work to different individuals. During the second meeting, the team discussed their completed parts and went over clarifications or potential changes. As the team leaders for Interim Project I, Quincy and Adam oversaw these meetings. After this meeting, the team remotely performed a practice run of their presentation. They performed more practice runs after the end of class on Tuesday, September 29th.

Introduction

The purpose of this project is to develop a smartphone app to help blind people navigate indoors safely. Specifically, the team is designing the application to help an unsighted person navigate the ECSS building of UTD. Although guide dogs, canes, and braille lettering on classroom doors are all beneficial tools for unsighted individuals, a smartphone app could take visual input and relay auditory instructions. Our smartphone application, named Theia for the Greek goddess of vision, could provide unsighted individuals with auditory instructions on the number of steps to take and where to turn when traversing a particular route. Users may choose different routes to account for factors like duration, presence of hazards, or potential foot traffic, allowing them to customize their experience.

2. Issues with the Preliminary Definitions Given

2.1 Issues with II.1 The Domain, Stakeholders, Functional and Non-Functional Objectives

DI1. "Indoors" is undefined

Preliminary definition	"For this smartphone app, the domain will be indoors, which can consist of multiple floors." Summary: "from one location in a building to another location in the same building or a different building."
Type	Ambiguity, incompleteness
Issue	The definition does not say which buildings or floors the app covers, or what makes two buildings "connected to each other."
Options	O1. Every UTD building. O2. Every floor of ECSS.

	O3. The ECSS 2nd floor only, expanding in Phase 2.
Decision	O3
Rationale	Dr. Chung stated that one floor is enough scope. One floor can be mapped in detail and every feature tested on it. Floors are stored as map data, so adding a floor or building later needs no new features.
Satisfied by	NFR5, NFR7

DI2. No map or positioning method

Preliminary definition	"Telling the user to stop at the right place to turn" and "use of as many sensors that are available on an advanced smartphone would likely lead to a more helpful smartphone app."
Type	Incompleteness (missing domain assumption)
Issue	GPS does not work reliably indoors. The definition never says where the building layout comes from or how the phone knows where the user is standing.
Options	O1. Bluetooth beacons installed along hallways. O2. Wi-Fi signal positioning. O3. The camera reads room-number signs against a preloaded floor plan. O4. Step counting from a known starting point.
Decision	O3 + O4. Step counting tracks the user between signs, and each sign the camera reads corrects the accumulated drift.
Rationale	Needs no hardware installed in the building and no approval from UTD Facilities, and uses only the phone's own sensors, as the definition encourages. Beacons were rejected because someone would have to install and maintain them.
Domain assumptions	An accurate floor plan of the mapped floor exists. Room-number signs are posted and readable by the camera.
Satisfied by	FR2, FR6

DI3. The caretaker's role is unclear

Preliminary definition	"A caretaker (an assistive person), e.g., a family member, who sets the configuration of the smartphone app and come to aid the blind person in case the blind person is lost or injured."

Type	Ambiguity
Issue	To "come to aid," the caretaker has to know where the user is. The definition does not say whether the user's location is shared at all times.
Options	O1. The caretaker does setup only. O2. Live location is shared at all times. O3. Location is shared only during an emergency or when the user asks for help.
Decision	O3
Rationale	The caretaker can still find the user when it matters, while the user keeps privacy and independence day to day. Consistent with NFRI4 (privacy).
Satisfied by	FR12, NFR8

DI4. "Blind" is not scoped

Preliminary definition	"Unsighted (i.e., blind) people" and "The primary stakeholder would be a blind person who needs to navigate indoors."
Type	Ambiguity
Issue	Blindness ranges from total blindness to low vision, and some blind users also have hearing loss. A voice-only app fails users who cannot hear it.
Options	O1. Totally blind users only. O2. Add low-vision support (large text, high contrast). O3. O2 plus a vibration-only mode.
Decision	O3, with distinct vibration patterns for the four core cues: stop, turn left, turn right, obstacle.
Rationale	Vibration is already needed in loud hallways (Scenario 2), so the added cost is small, and it widens who can use the app.
Satisfied by	FR11, NFR4, NFR6

2.2 Issues with II.2 Software System Requirements: Functional Requirements

FRI1. Emergency calls are underspecified

Preliminary definition	"Placing emergency calls and messages, possibly after detecting a fall or when the system cannot figure out the current location."

Type	Incompleteness, unsoundness
Issue	The requirement does not say who is called or what happens if the user cannot respond. Calling for emergency help only because the location is lost is out of proportion, since being lost is not an emergency.
Options	O1. Always call 911. O2. Always call the caretaker. O3. The user chooses by severity (caretaker, UTD Police, or 911). With no answer within 30 seconds, the system calls 911 and notifies the caretaker.
Decision	O3. When the location is lost, the system offers to contact the caretaker and never calls police.
Rationale	The user judges the severity, an unresponsive user still gets help, and false 911 calls are avoided. Shown in Scenario 3.
Satisfied by	FR8, FR8.1, FR8.2, FR9

FRI2. Obstacle detection is vague

Preliminary definition	"Detecting obstacles and telling the user what to do in order to avoid collision."
Type	Incompleteness
Issue	There is no definition of what counts as an obstacle or how far ahead the user is warned, and a phone kept in a pocket cannot see anything.
Options	O1. Floor-level obstacles only. O2. Anything in the walking path from floor to head height, including drop-offs such as stairs. O3. O2 plus moving people.
Decision	O2. The camera rides on a chest lanyard, warns at least 4 steps ahead, and gives a direction to move. Moving people are deferred to Phase 2.
Rationale	Head-height objects and drop-offs are exactly what a cane misses (Scenario 2), and the lanyard keeps both hands free. Predicting where people are moving is much harder, so it waits for Phase 2.
Satisfied by	FR7, FR11

FRI3. Distance units are ambiguous

Preliminary	"Telling the user to walk a distance (e.g., 2 minutes to reach a turning point, 30

definition	steps to take)."
Type	Ambiguity
Issue	The requirement mixes time and steps, and step length differs from person to person, so "30 steps" is not a fixed distance.
Options	O1. Steps. O2. Feet. O3. Seconds of walking. O4. The user's choice, with steps calibrated to the user's stride during setup.
Decision	O4, defaulting to steps.
Rationale	Steps are easy to count while walking, calibration makes them accurate, and the choice supports the customizable NFR.
Satisfied by	FR4, FR4.1, NFR6

FRI4. No starting-location or rerouting requirement; the list is open-ended

Preliminary definition	Summary: "a blind person may need to figure out what the source (starting) location is." The functional requirement list ends with "……."
Type	Incompleteness
Issue	No functional requirement determines where the user starts or handles the user leaving the route, and the open-ended "……." leaves the list incomplete.
Options	O1. The user states the starting room. O2. The system detects the start from the nearest room sign and the user confirms it. O3. O2 plus automatic rerouting when the user drifts off course.
Decision	O3. The list is closed so that every feature is a numbered functional requirement.
Rationale	Users often do not know which room they are next to, and step counting drifts (DI2), so rerouting is necessary.
Satisfied by	FR2, FR6

2.3 Issues with II.3 Software System Non-Functional Requirements

NFRI1. Safe, fast and comfortable conflict

Preliminary definition	"The system shall help the user safely navigate indoors; The system shall lead the user through the fastest route; The system shall lead the user through the route the user would feel the most comfortable with."
Type	Conflict
Issue	The three NFRs can each select a different route, and the definition does not say which takes priority.
Options	O1. Always the safest route. O2. The user ranks the three once in settings. O3. The safest route by default; saying "I'm running late" switches to the fastest route that still avoids known hazards.
Decision	O3
Rationale	Safety is the definition's top concern, lateness is the time concern it raises, and the user stays in control.
Satisfied by	FR13, NFR1, NFR2, NFR3

NFRI2. Mixed "shall" and "should"

Preliminary definition	Five NFRs use "shall"; "The system should be customizable" and "The system should be easily extensible" use "should."
Type	Inconsistency
Issue	"Should" makes these two NFRs optional, yet our decisions in DI4 and FRI3 depend on customization.
Options	O1. Keep "should." O2. Change both to "shall." O3. "Shall" for customizable only.
Decision	O2
Rationale	In Phase 1 every requirement uses "shall," and customization is required by our design.
Satisfied by	NFR6, NFR7

NFRI3. "Usable" and "ubiquitous" are vague

Preliminary	"The system shall be usable;" and "The system shall be ubiquitous;"

definition	
Type	Ambiguity
Issue	Neither statement can be tested. "Ubiquitous" could mean every building, every phone, or always online.
Options	O1. Leave them as general goals. O2. Define measurable versions.
Decision	O2. Usable: every function works by voice alone, the app works with VoiceOver and TalkBack, and a new user completes a first route after 10 minutes of training. Ubiquitous: the app runs on current iOS and Android versions and works offline on any mapped floor.
Rationale	Measurable NFRs can be tested. Offline operation matters because indoor signal is unreliable.
Satisfied by	NFR4, NFR5

NFRI4. Missing privacy and battery NFRs; two NFRs end in "etc."

Preliminary definition	The full II.3 list; the customizable NFR ("the volume, the interval..., etc.") and the extensible NFR ("new sensors and hardware etc.").
Type	Incompleteness
Issue	The app shares the user's location with a caretaker and police but has no privacy NFR. The camera and sensors run continuously, but there is no battery NFR. Two NFRs end in "etc."
Options	O1. Add a privacy NFR. O2. Add privacy and battery NFRs. O3. O2 plus replacing each "etc." with a complete list.
Decision	O3. Privacy: location leaves the phone only during an emergency or when the user asks (DI3). Battery: no more than 15% of battery per 30 minutes of navigation. Customizable: voice speed, volume, cue interval, distance unit, vibration strength and language. Extensible: new interfaces, languages, user needs, features, sensors and hardware.
Rationale	Closes the gaps our own features create, and a requirement never ends in "etc."
Satisfied by	FR12, NFR6, NFR7, NFR8, NFR9

Issues with II.1 The Domain, Stakeholders, Functional and Non-Functional Objectives

DI1
Description: "Indoors" is undefined: which buildings and floors, and what counts as "connected"?
Issue Type: *Ambiguity, incompleteness*
Options: Either store the ECSS floor plan as a single object or partition the data into different "phases."
Decision: Phase 1 covers the ECSS 2nd floor. Floors are stored as map data, so more can be added without new features.
Rationale: New phases can be added as needed, improving the scalability of the map data.

DI2
Description: No map or positioning method, and GPS does not work reliably indoors.
Issue Type: *Incompleteness (missing domain assumption)*
Options: Attempt to use GPS although it is unreliable, or use visual data to discern location.
Decision: The camera reads room-number signs against a preloaded floor plan; step counting tracks the user between signs.
Rationale: If the application already has camera permissions, it can use visual data to precisely pinpoint the user's location.

DI3
Description: The caretaker must "come to aid," but is location shared all the time?
Issue Type: *Ambiguity*
Options: Automatically share location whenever the application is running, ask to share location every time the application is opened, or only share location during emergencies.
Decision: Location is shared only during an emergency or when the user asks for help.
Rationale: The user does not need their location broadcasted every time they use the application. They only need to be located if they need manual assistance.

DI4
Description: "Blind" is not scoped: total blindness, low vision, blindness with hearing loss.
Issue Type: *Ambiguity*
Options: Consider the target audience as only those totally blind or consider a wide range of individuals with impaired sight.
Decision: Add low-vision support and a vibration-only mode with four distinct cues.
Rationale: Even if users are not totally blind, the application may still be helpful to them. Having Theia as an option creates redundancy with other aids like canes, allowing visually impaired users a wider variety of tools. If the user were to face some difficulty with their cane, then Theia

could serve as a backup option. In addition, offering different levels of support allows Theia to reach a much wider audience.

Issues with II.2 Software System Requirements: Functional Requirements

FRI1

Description: Emergency calls: nobody is named to call, and losing the location is treated as an emergency.
Issue Type: *Incompleteness, unsoundness*
Options: Default to 911, default to the UTD police, or implement a list that calls the user's chosen contacts in descending order.
Decision: The user picks caretaker, UTD Police or 911; no answer in 30 s calls 911 and notifies the caretaker. Being lost only offers the caretaker.
Rationale: By assigning responsibility to a single caretaker, the caretaker is more likely to take initiative and respond. If the user becomes lost within ECSS, they may choose to notify their caretaker, but this situation does not warrant notifying the UTD police or the 911 dispatch. In the event of an accident, the user is allowed to pick whether to notify their caretaker, the UTD police, or the 911 dispatch.

FRI2

Description: Obstacle detection: which obstacles, how far ahead, and how can a pocketed phone see?
Issue Type: *Incompleteness*
Options: Clearly define what types of obstacles should be recognized by the application. Either require the user to hold the phone in front of them or set up an apparatus that holds the phone for them.
Decision: Floor to head height, including drop-offs, at least 4 steps ahead, via a chest-lanyard camera. Moving people wait for Phase 2.
Rationale: If a user has to hold their phone in front of them the whole time, their arm may become fatigued. It may be difficult for them to hold the phone at a consistent height and position for a prolonged period of time, decreasing the quality of visual data sent to Theia and the usefulness of Theia's subsequent recommendations. Attaching the phone to a lanyard improves consistency and is more convenient for the user.

FRI3

Description: Distance mixes minutes and steps, and step length differs per person.
Issue Type: *Ambiguity*
Options: Default to standard units of measurement such as feet or meters, or calibrate route distance based on the user's steps.
Decision: The user picks steps, feet or seconds (default steps); stride is calibrated at setup.
Rationale: The app defaults to step calibration since this seems to be a more intuitive way of understanding remaining distance. Step calibration eliminates the need for users to calculate how their unique stride corresponds mathematically to standard units of measurement.

However, if the user would prefer the route to be displayed in units of measurement, the option is still available. The user may also select seconds if counting time is more intuitive for them than counting distance.

FRI4
Description: No FR finds the starting location or reroutes, and the list ends in "……."
Issue Type: *Incompleteness*
Options: Use GPS to guess the user's approximate location, use visual data to determine the user's starting location, or allow the user to input their location manually.
Decision: Detect the start from the nearest sign and confirm it; reroute when off course; every feature gets a numbered FR.
Rationale: It would be tedious for an unsighted individual to use the braille signage to input their starting location every time, and GPS can be inaccurate. The application is already using the camera, so it can determine the starting location using visual data before commencing the navigation process.

Issues with II.3 Software System Non-Functional Requirements

NFRI1
Description: Safe, fastest and most comfortable can each pick a different route.
Issue Type: *Conflict*
Options: Must default to either safest, fastest, or most comfortable, but should allow the user to manually switch to a different priority at any time.
Decision: Safest by default; saying "I'm running late" switches to the fastest route that still avoids known hazards.
Rationale: The purpose of the app is to improve safety for unsighted individuals, so the application defaults to the safest route. If the user is in a hurry, they may pick a faster route, but this route must still avoid hazards. The team defines a convenient route as one lacking high-traffic areas, and this route still must be hazard-free.

NFRI2
Description: Customizable and extensible say "should"; every other NFR says "shall."
Issue Type: *Inconsistency*
Options: Correct the original NFR or eliminate it; precise language is essential for requirements.
Decision: All NFRs use "shall." Our design depends on customization, so it cannot be optional.
Rationale: Changing the word from "should" to "shall" is a simple fix and allows the team to preserve the original NFR.

NFRI3
Description: "Usable" and "ubiquitous" cannot be tested as written.
Issue Type: *Ambiguity*

Options: Determine ways that the software can be made more useful for an unsighted individual. Try to make software supported by as many device types as possible, or focus on producing a high-quality application for a single device family.
Decision: Usable: every function by voice, works with VoiceOver and TalkBack, learnable in 10 minutes. Ubiquitous: iOS and Android, offline on mapped floors.
Rationale: Auditory output is a critical method for conveying information to an unsighted user, and VoiceOver and TalkBack are reputable screen readers that can be integrated into the application. At the current stage, the team believes it feasible to create prototypes compatible with both iOS and Android.

NFRI4
Description: No privacy or battery NFR, and two NFRs end in "etc."
Issue Type: *Incompleteness*
Options: Add an explicit privacy NFR, or assume that the scope of the application does not violate any Texas law. Add an explicit battery NFR, or test the application's performance throughout development and make changes as necessary.
Decision: Add privacy (NFR8) and battery at most 15% per 30 minutes (NFR9); replace each "etc." with a complete list.
Rationale: The team chose to define these NFRs upfront, to promote a more focused design and an increased ease of traceability.

WRS

World

Stakeholders – For, Of, By

The "of the people" stakeholder group consists of **caretakers**, **UTD police**, and the **911 dispatch**. Caretakers are responsible for responding to application alerts, and may be notified if the user is lost or is in an emergency. The UTD police are responsible for answering emergency calls on campus, and the 911 dispatch may act as backup if UTD's police force is not available. The user may choose to directly contact the UTD police or the 911 dispatch if there is an emergency.

The "by the people" stakeholder group consists of the project team, **Group 5**. The team is solely responsible for designing and building the application.

The "for the people" stakeholder group consists of **blind and low-vision users**, the **UTD Student AccessAbility office**, and **bystanders**. The app primarily serves blind and low-vision users, who will use it to help them navigate ECSS safely. As a disability aid, the application closely aligns with the goals of the UTD Student AccessAbility office, and the office may encourage the use of Theia. Since the application relies on the phone camera, it may observe several bystanders within the ECSS building as the user navigates the hallways.

Problem

Goal

Improved understanding of II.1 The Domain, Stakeholders, Functional and Non-Functional Objectives

"Try to clarify both the domain description and the
system description, while establishing the traceability between the two." - I think he means to just have separate tables for the domain, FRs, and NFRs, but the way this is worded confuses me.

ID	Improved Domain Description
ID1	For this smartphone app, the domain will be the ECSS first floor, including classrooms, offices, washrooms, lounges, elevators, stairs, and laboratories.
ID2	The app will use the existing floor plan of the building and room signs to determine the user's current position within the building.
ID3	The application will share the user's location with their caretaker only during an emergency, which occurs when the user has fallen or asks for help.
ID4	The primary stakeholders are blind and low-vision users, including users who are also deaf, who need to navigate indoors.
ID5	Secondary stakeholders will also include bystanders that will be seen by the camera during navigation. Any privacy laws/regulations concerning recording bystanders will need to be taken into account with regards to consent and data retention.

Requirements

Functional R – Improved understanding of II.2 Software System Requirements: FRs

VERBATIM FROM THE PROJECT1 PDF A problem with all of these requirements is that they don't follow the "The system shall…" model of writing functional requirements

1. Accepting from the user the destination location to go (The system might even suggest or confirm a possible destination location, utilizing the user's routine schedule or habit). This FR could be separated into 2. One FR for the user selection and a second FR for creating the list

2. Figuring out and telling the user which routes can reach the destination location, and accepting the user's preference Replacing "Figuring out…" with "Listing routes(s)"

3. Telling the user to walk a distance (e.g., 2 minutes to reach a turning point, 30 steps to take)

4. Telling the user to stop at the right place to turn Possible FR problem is that the "right place" is vague and unverifiable.

5. Detecting obstacles and telling the user what to do in order to avoid collision This could be broken up into multiple FR's. One encapsulating when to tell the user to stop and another handling detecting obstacles

6. Placing emergency calls and messages, possibly after detecting a fall or when the system cannot figure out the current location Specify what emergency calls and messages are made, like "Emergency services" or "User stated emergency contact". Removing "possibly" would improve it as well due to it being ambiguous.

7. Figuring out what would be the next action(s), based on the user's schedule or habit, and suggesting and accepting the user's choice) Replacing "Figuring out…" with "Listing action(s)"

Non-functional R -Improved understanding of II.2 Software System Requirements: NFRs

VERBATIM FROM THE PROJECT1 PDF
The system shall help the user safely navigate indoors;
The system shall lead the user through the fastest route;
The system shall lead the user through the route the user would feel the most comfortable with;
*does this imply different styles of routes or routes differing depending on the person?
The system shall be usable; without sight. Using only haptics and audio (part of this can spawn FR's)
The system shall be ubiquitous; this could be a bad requirement as ubiquitous seems as inclusive to me as perfect
The system should be customizable to every user (e.g., the volume, the interval at which the system says something, the order whereby different things the system says, etc.)
The system should be easily extensible to accommodate the following typical variations: variations in interface, language, definitive needs of the user, new features, new sensors and hardware etc.

Specification

Functional S – Software functional specification, from II.2 Software System FRs

Non-functional S –Software non-functional specification, from II.2 Software Sys NFRs

Prototype and User Manual



Home
Voice-first: the whole screen is one button

Guidance
Step countdown in the user's chosen unit

Obstacle
Full-screen warning plus rapid vibration

Fall check
Choose who to call; 911 after 30 s

High contrast, huge type and 64 pt tap targets for low-vision users (DI4, NFR4). Every screen is also spoken.

Quick start

1. Open Guide Dogs. Every screen is spoken, and VoiceOver and TalkBack work.
2. Choose a distance unit: steps, feet or seconds.
3. Calibrate your stride by walking 10 normal steps.
4. Practice the four vibration cues.
5. Add your caretaker as an emergency contact (a sighted helper can do this).
6. Clip the phone to the chest lanyard, camera facing forward.
7. Tap anywhere and say where to go: "Take me to ECSS 2.410."

Your location is shared only during an emergency or when you ask for help.

Vibration	Meaning
One long buzz	Stop
Two short	Turn left
Three short	Turn right
Rapid pulses until clear	Obstacle ahead

Command card

Say	What happens
"Take me to [room]"	Starts a route
"Where am I?"	Tells you the nearest room
"What are my options?"	Lists the available routes
"Repeat"	Repeats the last instruction
"I'm running late"	Fastest hazard-free route
"Pause" / "Stop navigation"	Pauses or ends the route
"Help"	Offers to contact your caretaker
"I'm fine"	Cancels a fall alert
"Call caretaker" / "Call campus police" / "Call 911"	Places that call and sends your location

Phase 1 draft; caretaker setup screens come in Phase 2.

Traceability

Trace ID	Need / Finding	Domain Assumption	Functional Req (FR)	Non-Functional (NFR)	System Feature
TR-01	Hands-free way to specify destinations without touch screens	User speaks clearly; mic is unobstructed	**FR1 / FR3:** Voice destination input and stride-calibrated guidance	**NFR3 / NFR4:** Voice-first usability, high contrast, ≤10 min learning curve	**Home & Guidance:** Single full-screen tap-to-talk
TR-02	Cane misses elevated/hanging obstacles (head height)	Forward-facing chest lanyard camera, lit hallways	**FR7 & FR11:** Object detection within 4 steps + haptic alerts	**NFR1:** Safe navigation, hazard-free routing	**Obstacle Screen:** Yellow warning + rapid pulses
TR-03	User falls in empty hallway and phone flies out of reach	Accelerometer detects impact & lack of movement	**FR6/8/12:** Fall check-in, 30s countdown, auto-dial emergency	**NFR8 / NFR1:** Safety & privacy (emergency location only)	**Fall Alert Screen:** Red 30s auto-escalation screen
TR-04	Battery drainage leaves blind users stranded	Sensors optimize duty cycles during steady paths	**FR8:** Proactive voice alerts at low battery (20%, 5%)	**NFR9:** Power consumption ≤ 5% per hour	**System Service:** Background battery guard daemon

Reference

Appendix

Index